IL-TUR: Benchmark for Indian Legal Text Understanding and Reasoning

Joshi, Paul, Sharma, Goyal, Ghosh and Modi (ACL 2024), arXiv:2407.05399

Paper summary for the CS-781 capstone

The paper at a glance

Citation

A. Joshi, S. Paul, A. Sharma, P. Goyal, S. Ghosh, A. Modi. *IL-TUR: Benchmark for Indian Legal Text Understanding and Reasoning*. ACL 2024 (arXiv v2, November 2024). IIT Kanpur and IIT Kharagpur.

One-sentence summary

A single benchmark of eight Indian legal NLP tasks, with unified data formats, baselines (including LLMs) and a public leaderboard. Current models are far from solving it.

Why it matters for our capstone

This is the capstone's **base paper**. Phases 2–7 are IL-TUR tasks: L-NER, RR, CJPE, LSI, SUMM and PCR.

Where it appears in our slides

Main deck: "Capstone at a glance" and every Phase 2–7 slide.

Roadmap

1. The problem
2. The benchmark
3. Results
4. Assessment and outlook

Why legal NLP for India?

- India has about **50 million pending cases** across district, state and federal courts (National Judicial Data Grid, 2023).
- "Justice delayed is justice denied."
- Legal work runs on documents, so NLP could help lawyers and judges read, search and summarise faster.
- But legal NLP tasks existed in isolation, with different formats and metrics, so progress was hard to track.

The idea in one line

Do for Indian legal NLP what GLUE did for general NLP: one benchmark, one format, one leaderboard.

Not meant to replace judges

The goal is to *augment* legal practitioners, e.g. by suggesting statutes, surfacing precedents or summarising, not to automate decisions.

Why a separate benchmark for legal text?

Challenge	What it means, with an example
1. Domain vocabulary	Everyday words take special meanings: "consideration", "relief", "ratio", "stay"
2. Very long documents	An Indian Supreme Court judgment averages about **4,000 words**; BERT reads only 512 tokens
3. Unstructured and noisy	Many documents are typed manually in courts, with typos and no consistent sections
4. Sub-domains differ	A model trained on civil law may fail on banking or insurance law: domain adaptation is hard
5. Explainability required	A lawyer or judge must see *why* a model predicts an outcome

Mixed legal system

India is mainly **common law**: higher courts can overrule precedents, so outcomes need reasoning beyond the statute book. Some areas (e.g. banking, insurance) follow **civil law**. IL-TUR covers both.

How IL-TUR differs from other legal benchmarks

Benchmark	Jurisdiction	Task types	Languages
LexGLUE	US, EU	classification	English
LEXTREME	EU, Brazil	classification	24 EU
FAIRLEX	EU, US, China, CH	fairness	several
LBOX	Korea	class., generation	Korean
LegalBench	mostly US	class., generation	English
LawBench	China	class., gen., extraction	Chinese
IL-TUR	**India**	**class., retrieval, gen., extraction**	**English + 9 Indian**

Key differences

Tasks are grounded in the **real legal workflow**, not exam-style questions. Documents are **long**: about 4,000 words, against about 200 in LegalBench. Tasks include retrieval, generation and explanation, and both common and civil law are covered.

Design principles

1. **Legal understanding and world knowledge**: tasks need real legal knowledge plus common sense.
2. **Difficulty**: not solvable by a layperson; hard for current models.
3. **Language**: English and the most frequent Indian court languages.
4. **Automatic evaluation** with metrics that align with human judgement.
5. **Public availability**: CC BY-NC-SA data and a public leaderboard.

Harmonisation

Several tasks existed before, each with its own format and metric. IL-TUR converts all of them to a **uniform JSON format**, collects training scripts, and defines a standard evaluation.

Anonymisation

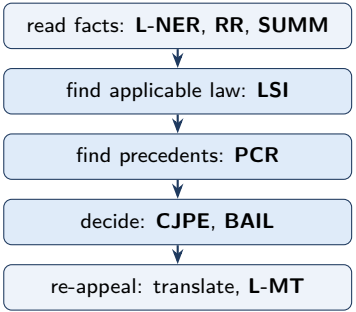
Named entities are anonymised in RR, CJPE, BAIL, LSI and PCR data, to address ethical concerns and to stop models from learning biases tied to names.

The eight tasks

Task	Dataset	Avg words	Type	Key skill
L-NER	105 docs, 650k words	6,180	sequence labelling	legal roles of entities (new task)
RR	21,184 sentences	25,796	multi-class	legal semantics per sentence
CJPE	ILDC, 34k docs	3,336	class. + extraction	understanding and reasoning, with explanation
BAIL	HLDC, 176k docs (Hindi)	86	classification	reasoning in Hindi
LSI	ILSI, 65k samples	2,406	multi-label	applying statutes to facts
PCR	IL-PCR, 7,070 docs	8,096	retrieval	legal relevance between cases
SUMM	In-Abs, 7,130 docs	4,376	generation	legal understanding and writing
L-MT	MILPaC, 17,853 pairs	49	generation	English $\leftrightarrow$ 9 Indian languages

Note the lengths: prior case retrieval documents average about 8,100 words, far beyond a 1,024-token window.

Tasks in the litigation process, and in our capstone



Capstone phase	IL-TUR task
Phase 2	L-NER
Phase 3	RR (CL and IT domains)
Phase 4	CJPE (prediction part)
Phase 5	LSI
Phase 6	SUMM
Phase 7	PCR

BAIL (Hindi) and L-MT (translation) are not capstone phases. Phase 1, the foundation model, is the capstone's own addition.

Easy examples of each capstone task

Task	Input (simplified)	Output
L-NER	"Appeal by *Ram Kumar* against the order of the *High Court of Delhi* dated *3 May 2019*..."	APP, COURT, DATE spans
RR	"The accused was arrested on 3 May." / "Whether Section 302 applies."	Fact / Issue
CJPE	full judgment with the final order removed	1 (appeal accepted)
LSI	facts: "the accused, together with two others, planned and caused the death of..."	IPC 302, IPC 120B
PCR	query judgment with citations removed, and a pool of candidate judgments	top-7 relevant cases
SUMM	4,000-word judgment	a few-paragraph summary

Why L-NER is not ordinary NER

A standard NER model tags the judge, petitioner and respondent all as PERSON. L-NER must know their *legal roles*, which needs legal understanding. Its gold data was annotated by law students, with entity definitions from legal academics.

Best baselines per task

Task	Best score	Metric	Model
L-NER	48.58%	strict macro-F1	InLegalBERT + CRF
RR	69.01%	macro-F1	MTL-BERT
CJPE	81.31%	macro-F1	InLegalBERT + BiLSTM
BAIL	81%	macro-F1	TF-IDF + IndicBERT
LSI	28.08%	macro-F1	LeSICiN (graph-based)
PCR	39.15%	μF1@K	event-based model
SUMM	0.33 / 0.86	ROUGE-L / BERTScore	Legal-LED
L-MT	0.28 / 0.32 / 0.57	BLEU / GLEU / chrF++	Microsoft Translator

Reading it

Strict L-NER is below 50% (exact boundaries are hard), LSI below 30% macro-F1 (rare statutes), and PCR below 40% (legal relevance is subtle). These are the scores our Phase 2–7 systems will be compared against.

Does in-domain pre-training help?

Task	Setup	BERT	LegalBERT	InLegalBERT
L-NER	flat, chunks	39.59	45.58	**48.58**
RR	flat, sentences	58	54	58
CJPE	hierarchical	71.14	78.21	**81.31**
LSI	hierarchical	18.44	21.74	**26.23**
PCR	flat, chunks	**9.24**	8.67	7.57

Macro-F1 (strict for L-NER). LegalBERT: pre-trained on EU/US legal text. InLegalBERT: on Indian legal text.

Lesson for our Phase 1

Except for PCR, performance rises from general BERT, to foreign legal BERT, to **Indian** legal BERT: it tracks how in-domain the pre-training was. This supports our plan to pre-train from scratch on Indian legal text. For CJPE, *hierarchical* models (encoder + BiLSTM over chunks) worked best, because judgments exceed 512 tokens.

Large language models are far behind

Best GPT-3.5 / GPT-4 score over 0-, 1- and 2-shot prompts, vs the task-specific best:

Task (metric)	GPT-3.5	GPT-4	Best baseline
L-NER (strict mF1)	32.84	24.03	**48.58**
RR (mF1)	30.95	38.18	**69.01**
CJPE (mF1)	56.74	68.29	**81.31**
BAIL (mF1)	61.0	66.67	**81**
LSI (mF1)	22.61	23.99	**28.08**
SUMM (ROUGE-L)	0.22	0.23	**0.33**
L-MT (BLEU)	0.26	**0.36**	0.28

Reading it

- Task-specific models beat GPT-4 on every task except translation.
- Long contexts and structured outputs (spans, one label per sentence) are hard to get from prompting.
- Extra examples did not reliably help; one-shot was sometimes worse than zero-shot.
- PCR could not be tried: the candidate pool exceeds even GPT-4's context.

Generalising across jurisdictions

Most tasks are jurisdiction-specific (statutes, bail practice), but summarisation can be tested across countries: train on Indian **In-Abs**, then test zero-shot on UK Supreme Court summaries (**UK-Abs**).

Model	Trained on	ROUGE-1	ROUGE-L	BERTScore
SummaRuNNer	In-Abs	**0.604**	0.225	0.828
	UK-Abs	0.493	**0.274**	**0.849**
Legal-LED	In-Abs	**0.557**	0.242	0.844
	UK-Abs	0.482	**0.264**	**0.851**

Reading it

Models trained only on Indian data summarise UK judgments decently: higher ROUGE-1, and comparable ROUGE-L and BERTScore. Cross-jurisdiction adaptation is left as future work.

Strengths and limitations

Strengths

- First unified benchmark for Indian legal NLP.
- Realistic, workflow-grounded tasks with long documents.
- Several new tasks and datasets (e.g. L-NER).
- Uniform formats, public data and leaderboard.
- Strong baselines, including LLM comparisons.
- Anonymisation to reduce bias.

Limitations (mostly stated by the authors)

- Not every legal sub-domain is covered in each task.
- Multilingual only for L-MT; BAIL only in Hindi.
- Some datasets were built with ML tools (e.g. fact extraction for LSI), so labels may be imperfect.
- Explainability is addressed mainly in CJPE.
- LLMs were evaluated on subsets, because of cost.
- Only eight tasks so far.

Future research directions

Stated by the authors

- Add tasks, e.g. **legal coreference resolution** for information extraction and knowledge graphs.
- A **single model for all tasks**, e.g. mT5 or multi-task adapters.
- More Indian languages and lower-court data.
- Refined datasets built with legal professionals.
- Cross-jurisdiction domain adaptation.
- Two-stage retrieval so LLMs can attempt PCR.

Directions our capstone explores

- A **self-hosted foundation model** trained only on Indian legal text and reused across six tasks, close to the authors' "one model for all tasks" goal.
- Parameter-efficient adaptation (LoRA, LORMA) and distillation for small, private deployments.
- Long-document strategies (chunking, hierarchies, ALiBi or RoPE) for CJPE, LSI and PCR.

Summary

- India's court backlog motivates legal NLP; legal text is long, noisy, specialised and needs explanations.
- **IL-TUR**: eight tasks (L-NER, RR, CJPE, BAIL, LSI, PCR, SUMM, L-MT), English and 9 Indian languages, unified formats and a public leaderboard.
- Best baselines leave large gaps: strict L-NER 48.6%, LSI 28.1%, PCR 39.2%.
- In-domain (Indian legal) pre-training helps on most tasks; hierarchical models help on long documents.
- GPT-3.5 and GPT-4 trail task-specific models on every task except translation.
- For us: Phases 2–7 *are* IL-TUR tasks, and these baselines are the bar to beat.

Reference

Joshi, A., Paul, S., Sharma, A., Goyal, P., Ghosh, S., and Modi, A. (2024). IL-TUR: Benchmark for Indian Legal Text Understanding and Reasoning. In *Proceedings of ACL 2024*. arXiv:2407.05399. Leaderboard: `https://exploration-lab.github.io/IL-TUR/`.

Other work cited: Malik et al. (2021) ILDC; Malik et al. (2022) Semantic Segmentation of Legal Documents; Kapoor et al. (2022) HLDC; Paul et al. (2022) LeSICiN; Joshi et al. (2023) U-CREAT (event-based PCR); Paul et al. (2023) InLegalBERT; Chalkidis et al. (2022) LexGLUE; Guha et al. (2023) LegalBench.